

# *Section 4: Architectural Patterns & Styles*

## Introduction

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Multi-tier architectural pattern is a fundamental architectural pattern that is used in modern software design. In this session we will explore how multi-tier architecture works, its key components and why it is widely used in scalable and secure applications. We'll break down the difference between two-tier, three-tier and n-tier architecture. We'll examine their impact performance and scalability and look at real world examples like web applications and enterprise systems.

By the end of this session, you will have a solid understanding of multi-tier architecture and how to apply it in system design.

## What is Multi-Tier Architecture?

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Multi-tier architecture is a software design pattern that structures an application into multiple layers, where each of these layers has a specific role. Instead of a monolithic system where everything is tightly coupled, multi-layer design separates the concerns into independent layers. This separation improves scalability, maintainability and security making it a popular choice for modern applications.

There are few points to remember when it comes to multi-tier architecture.

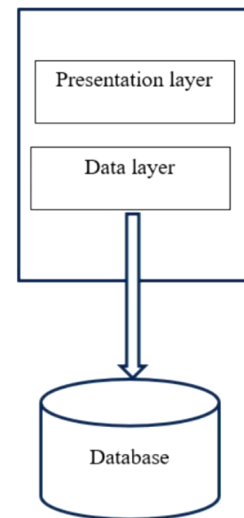
- It organizes applications into independent layers so changes in one layer don't affect the other layers so changes in one layer don't affect the other layers.
- It separates concerns. The UI, business logic and data storage layers are handled separately which improves the code organization and reusability.
- It enhances performance and security by allowing load balancing, caching and secure database access.
- It is widely used in web applications, enterprise systems and cloud architectures where scalability and reliability are crucial.

## 2-Tier Architecture

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In a 2-tier architecture we have 2 layers, the client layer and the database layer. The client layer typically represents the user interface or the application logic where the user interacts with the system. The database layer is where the application stores and retrieves data from. Usually through a SQL database or NoSQL database setup we have the data layer.

In this setup the client communicates directly with the database without any intermediate business logic layer. So the client layer is where the user interface and application logic reside. Example a desktop application where the user will input the data and interact with the application and the database layer will be the layer that will manage the data storage and retrieval. In this two-tier application the data flows directly between client and the database with no intermediate layers in between.



### Pros:

- It is very simple to implement as there are only two layers making it easy to set up and understand.
- It is very fast for small-scale applications. It is ideal for applications that don't require much scalability.

### Cons:

- It is very poor in terms of scalability. As the number of users grows, performance can degrade because there is no intermediary to handle business logic or distribute the load.
- There is also a security risk. Direct database access means that there are potential vulnerabilities as there is no business logic layer to out or secure the data.

### Usecase:

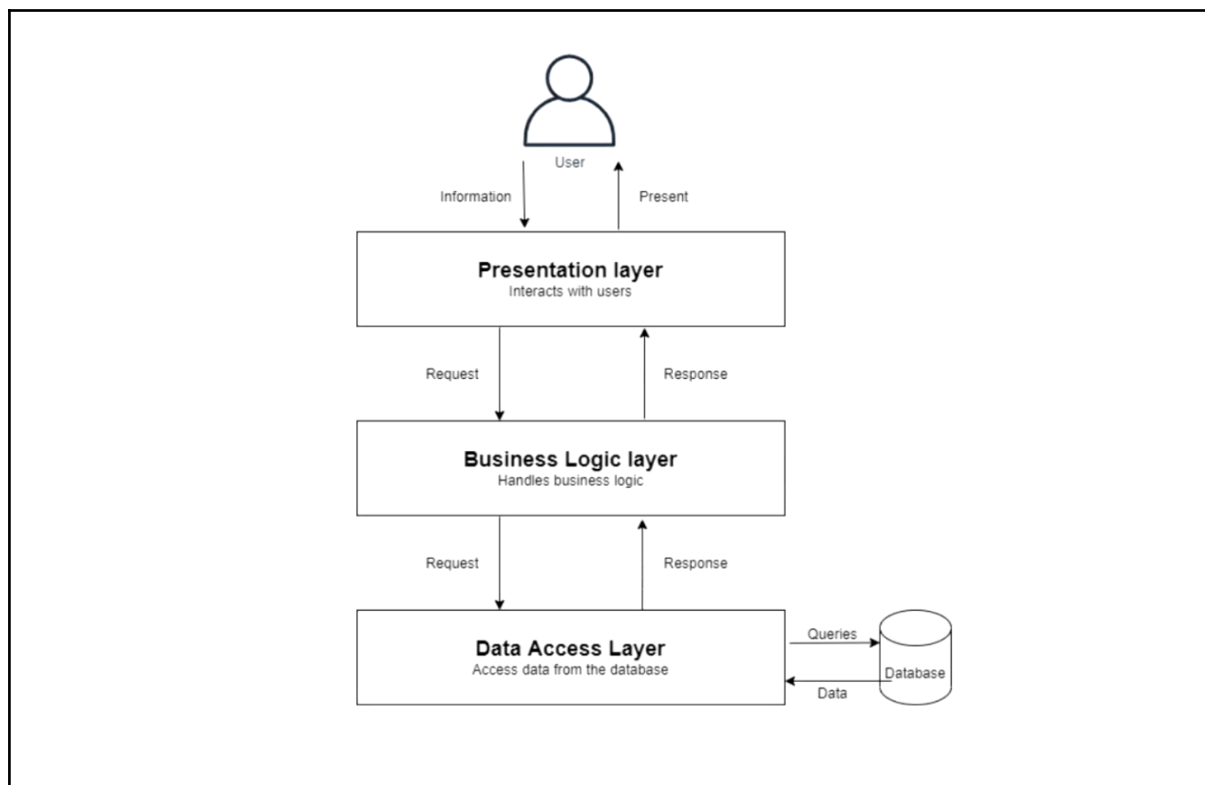
- The use case for such an architecture could be a desktop application that directly queries a SQL database that resides on the same machine. This is okay because scalability is not a primary concern here.

So while simple and effective for small scale applications, two-tier architecture is often unsuitable for larger more complex applications.

## 3-Tier Architecture

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The 3-tier architecture introduces an important middle layer which is known as a business logic layer. In a 3-tier architecture we have 3 layers, the frontend which will have the UI, the business logic layer and the database layer. This setup adds a separation of concern between the user interface, the application logic and the data.



So how does it work?

1. First we have the frontend layer. The user interacts with the business logic layer through an API. So the frontend will talk to the business logic layer.
2. The business logic layer will then process business rules, application logic and requests from the UI. It communicates with the database to fetch, update and delete the data.
3. Then we have the database layer, which stores and manages the application data. It only communicates with the business logic layer and not the UI layer directly.

So typically what happens is that we have a presentation layer which has the responsibility of all the UI related things, business logic layer where all my business and application logic will be there and finally the data access layer which will have all the data access related code and the data separately stored in the database. So unlike the two-tier architecture where the clients communicate directly with the database, the three-tier architecture uses a middle layer to handle the logic and secure database interactions.

**Pros:**

- Improves scalability and security. By separating the business logic layer we can independently scale each layer and enforce stricter security policies between the UI and the database.
- Better separation of concern. Each layer handles a specific responsibility making the system easier to manage and evolve over time.
- It is easier to maintain. Changes in one layer do not affect other layers making the system easier to maintain and extend.

**Cons:**

- There is slightly higher latency because we have an extra layer and adding an extra layer of processing introduces some latency as requests now go through the business logic layer before reaching the database.

**Usecase:**

- Traditional web applications are classic examples of three-tier architectures where it can have a front-end like a React or Angular application, communicate with the backend API which could be a business logic layer, which will talk to the data access layer, process the request and interact with the database to retrieve and store the data.

So to summarize, 3-tier architectures are highly suitable for applications that need better scalability, security and separation of concerns. It provides a solid foundation for modern web applications.

# N-Tier Architecture

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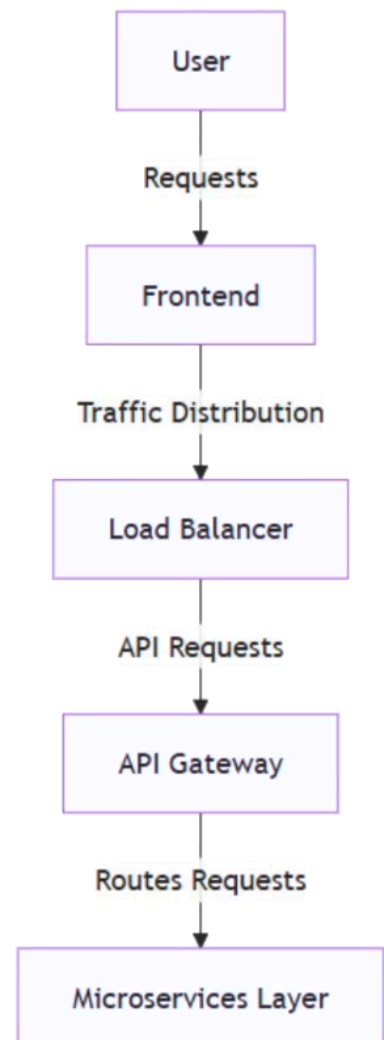
N-tier architecture extends the concept of three-tier by adding more specialized layers to handle more complex systems. In n-tier architecture we go beyond basic 3 layers by introducing additional specialized layers such as caching, API gateway, microservices and more. These extra layers help handle the complexity of modern high traffic applications.

N-tier architecture handles high traffic and complex business logic applications much better. The additional layer allows the system to distribute loads efficiently and process complex business logic in a scalable way. For example, caching layers can reduce database load and microservices can isolate functionality to keep things manageable. Second, independent scaling of services is also beneficial. Each layer can be scaled independently to meet specific needs. For instance, the caching layer can be scaled separately from the business logic layer allowing you to allocate resources based on demand. This makes the system more flexible and efficient in handling varied workloads.

Examples:

- Microservices-based applications often use n-tier architecture where each micro service is a separate tier on the layer in architecture allowing each service to scale and evolve independently.
- Large-scale enterprise software (E-commerce platforms, banking systems, CRM tools) also uses n-tier to scale and handle the traffic with specialized layers for things like security, caching, monitoring and analytics.

So in short, n-tier architecture is ideal for systems that require high scalability, fault tolerance and the ability to manage complex business logic. Its widely used in enterprise applications and large-scale services like social media platforms, e-commerce sites and cloud-native applications.



# Performance & Scalability Impacts

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In this section, we are going to discuss how the number of tiers in the architecture impacts system performance and scalability. So as we introduce more layers and more complexity, each layer adds a certain processing overhead which can affect performance and scalability if we are not managing it carefully.

- **Latency Considerations:** One key factor is when we are adding more tiers, then latency will increase. Every additional layer introduces more communication between different components which can lead to higher latency in data processing. That means the request must pass through multiple points which will introduce delays. So the more layers you have, the higher the latency unless those layers are more properly utilized. The below are some of the existing solutions to optimize latency:
  - **Caching (Redis, Memcached):** It is the way to optimize latency and improve performance. For instance, frequently accessed data can be stored in a cache so that it doesn't have to be received from the database every time. This can significantly reduce latency and improve response times.
- **Scaling Strategies:** We can scale our application to actually make it perform better in an n-tier architecture. So there are 2 types of scaling.
  - **Vertical Scaling:** This type of scaling involves adding more resources like CPU, RAM or storage to a single server. While this can help handle increased load for a while, it has limits. Eventually a single machine will reach its capacity. However for an application that doesn't require massive scale, vertical scaling can be a simple solution to improve the performance.
  - **Horizontal Scaling:** This type of scaling is more powerful for handling high traffic. It involves adding more servers to distribute the load and improve the fault tolerance. As the demand grows, you can simply add more machines allowing the system to handle large traffic volumes. This is especially useful for cloud-based systems that can scale on demand.

So to summarize, the more tiers you introduce in your system in an n-tier architecture, the greater the importance of optimizing the latency and scalability. Using techniques like caching and load balancing combined with the vertical and horizontal scaling can help you build a system that handles high traffic efficiently and provides low latency responses to the end user.

## Summary & Key Takeaways

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- ☒ Multi-Tier = Better Scalability & Maintainability.
- ☒ 2-Tier is simple but limited.
- ☒ 3-Tier is the standard for web apps.
- ☒ N-Tier is for large-scale, cloud-native systems.
- What's Next:
  - Microservices Architecture

## Interview Questions

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- Basic Questions:
  - What is Multi-Tier Architecture, and why is it used?
  - How does a 2-Tier architecture differ from a 3-Tier architecture?
  - What are the key components of a 3-Tier architecture?
  - Can you give an example of a real-world application that follows an N-Tier architecture?
- Intermediate Questions:
  - How does scalability improve in a multi-tier system?
  - What are the challenges of adding more tiers in an application?
  - How does load balancing work in an N-Tier architecture?
  - What strategies can be used to reduce latency in a multi-tier application?
- Advanced Questions:
  - In a large-scale system, when would you choose microservices over a traditional N-Tier architecture?
  - How would you secure inter-tier communication in a distributed multi-tier system?
  - How would you design a fault-tolerant multi-tier system for a high-traffic application like Netflix or Amazon?